Hypothesis Testing: The Steps

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MATH 240

Week 9, Lecture #1 — Ch. 11

Leading to the Exam

1. 10/26-30
   - Hypothesis testing procedures; sampling distributions
   - To-do: project proposal by 10/30
2. 11/2-6
   - Confidence intervals with bootstrapping; inference
   - To-do: HW #6 and Lab #3 by 11/2, CP4 on 11/6, HW #7 by 11/6
3. 11/9-13
   - Inference and decision errors
   - To-do: HW #8 by 11/13
   - **Friday, 11/13**: Exam #2

Feedback from Mid-Semester Survey

- Slides and R have been useful (despite challenges)
  - Intro to R Textbook
  - R for Data Science

Steps of Hypothesis Testing

1. Define the problem
   - Parameter of interest
   - Null and alternative hypotheses
   - Level of significance
2. Collect data or run experiment
   - Determine sample statistic
3. Choose a model for testing
   - Appropriate test/model
   - Verifying conditions
4. Perform test/model
   - e.g. test statistic to get p-values and/or confidence intervals
   - e.g. randomization testing
5. Draw conclusions
   - Make decision on null/alternative hypotheses
   - Interpret decision in context of the problem

A Step Back: Rolling a Die

Last lecture, we were curious about whether the dice I own are actually fair, such that the chance of rolling a 4 is $p = \frac{1}{6}$.

1. Define the problem:
   - Parameter of interest: p (true population proportion of rolling a 4)
   - Null and alternative hypotheses:

$$H_0 : p = \frac{1}{6}$$
$$H_a : p \neq \frac{1}{6}$$

Note: I care about having fair dice. Hence, I'm looking at both directions for my alternative hypothesis.

A Step Back: Rolling a Die

Last lecture, we were curious about whether the dice I own are actually fair, such that the chance of rolling a 4 is $p = \frac{1}{6}$.

1. Define the problem:
   - Significance/discernible level α: 0.05
     - ★ 5% chance that we incorrectly reject H_0

A Step Back: Rolling a Die

2. Collect the data:
   - We roll one die 24 times and find the number of times we roll a 4
   - This is our sample statistic $\hat{p}$

A Step Back: Rolling a Die

3. Choose a model for testing and perform it:
   - Used to establish baseline from null distribution
   - e.g. having you all roll the dice (for randomization testing)

Components of a Test

We need the following key components to finish these steps:

- **Test statistic**: a measure that summarizes how compatible the data are with H_0
  - Our test statistic is the sample proportion I got: $\hat{p} = \frac{8}{24}$
- **Sampling distribution**: distribution of a statistic across all samples
  - e.g. the distribution we created with the post-it notes (assuming we had more samples)
  - **Null distribution**: sampling distribution for the chosen test statistic

Components of a Test

We can then use our test statistic and compare to the null distribution, using the following procedures:

- **p-value**: the probability of obtaining a test statistic as rare or more rare than our observed test statistic if the null hypothesis were true
- **Confidence interval**: plausible range of values for the population parameter
  - Centered around our test statistic
- **Critical value**: value of test statistic(s) that are a threshold corresponding to boundary of rejection region

Note: We will focus on the p-value for now!

A Step Back: Rolling a Die

Assuming we took $B = 500$ samples, we can do what is called a **bootstrap**.

- Take B samples, in which we roll a die 24 times and calculate the proportion of 4s rolled
- Compute the p-value by taking proportion of B samples that produced proportions as or more extreme than our test proportion
  - We find that 65.8% of samples have proportionns as or more extreme

A Step Back: Rolling a Die

5. Draw conclusions:
   - Need to reject or fail to reject H_0
     - ★ Reject H_0 if p-value $\leq \alpha$
     - ★ Fail to reject H_0 otherwise
   - State conclusion in context

Since p-value $= 0.658 > 0.05 = \alpha$, we **fail to reject** H_0 and say that there is not enough discernible evidence to say that my die is different from any other dice.

A Few Notes

- We are never proving results with a hypothesis testing framework
  - There is never certainty
- Models and tests can vary depending on the parameter of interest and tools available
  - Nonparametric procedures (bootstrapping, randomization)
  - Classical techniques (z-test, t-test, F-test)
- Different ways to summarize information
  - e.g. p-values, confidence intervals, rejection regions

Let's Explore!

- Download the .qmd file from the website
- We will walk through hypothesis testing in R